Data-Driven Fault Detection in Rolling Element Bearings Using Machine Learning for Predictive Maintenance

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Abstract. Rolling element bearings are very important parts of rotating machines, but they often fail and cause problems. If a bearing breaks suddenly, it can stop the whole machine and lead to major losses. Due to this, it is necessary to monitor their condition regularly and detect faults at an early stage. Previous maintenance methods, like doing checks on a fixed schedule or manual inspection, are not always effective because they usually catch the problem too late. Nowadays, machine learning has become a useful tool for this task. By studying vibration signals, machine learning can find hidden patterns and help in detecting faults at an early stage. In our case we observed that supervised learning models and ensemble methods can give good results for fault diagnosis in different cases. Probabilistic methods also show that it is important to deal with uncertainty, especially when data is noisy or limited. In this study, a framework is proposed that uses statistical feature extraction along with adaptive classifiers for detecting bearing faults. The method is tested on benchmark datasets and shows good performance in classifying fault type and severity. The results show that data-driven machine learning can support predictive maintenance by increasing efficiency, reducing downtime, and improving reliability industries.

Keywords: Rolling Element Bearings, Condition Monitoring, Fault Detection, Machine Learning, Data-Driven Modeling, Predictive Maintenance.

1 Introduction

Rolling element bearings are one of the main parts of rotating machines because they support motion and carry loads with minimal friction. A fault in a bearing can act on the smooth operation of the system and may lead to unexpected breakdowns or high maintenance costs. To avoid this issue, it is essential to monitor the bearing condition regularly and find any fault at an early stage. Traditional condition monitoring methods mostly rely on manual inspection or fixed threshold techniques, which are time-consuming and often less accurate. Subsequent paragraphs, however, are indented.

Machine learning has become a powerful technique for automatic fault detection and health prediction. With the growth of data-driven methods, it can now analyze bearing health conditions more effectively and accurately. Machine learning has the ability to

handle large amounts of data, recognize small variations in signal features, and make reliable predictions without human interpretation. These methods can also adjust to different operating conditions and help in making timely maintenance decisions.

In this study, we use a data-driven approach to identify faults in an SKF 6206 deep groove ball bearing. The Root Mean Square (RMS) value of vibration signals is taken as the main feature for training and testing different machine learning algorithms such as K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Linear Regression, and Random Forest. The accuracy of each algorithm is compared to check its efficiency in classifying bearing health conditions. The aim of the experimental work is to find the most suitable algorithm for fault detection and to support predictive maintenance practices.

This research highlights how machine learning techniques can contribute to developing intelligent, data-driven maintenance systems. The proposed approach not only improves fault detection accuracy but also supports reliable decision-making for industrial applications where continuous monitoring is essential.

2 Literature Review

Hybrid feature selection was purposed by Tu et al. [1] combining a fisher score forward selection for SVM based bearing fault diagnosis by using real-time vibrational data from journal bearings under muddy water lubrication without artificial faults and found that the mean RMS value was the most significant feature which is capable of diagnosing faults accurately with 97 to 100% accuracy which proves that the is method is highly reliable.

A comparative study on bearing fault diagnosis was conducted by Chandrvanshi et al. [2], by using learning models like SVM, KNN, Linear and Ridge Regression, XGboost, CatBoost, etc. In this study; the bearing vibration data collected under different load conditions and processed through EDA, training and testing parts and compared each model's accuracy to know which is the best performing algorithm for fault detection of bearing.

A Random Forest (RF) based approach was proposed by Patel et al. [3], for multi class fault diagnosis in induction by extracting statistical features like RMS, etc. From vibration signals and compared RF performance with neural networks classifiers, which shows higher accuracy and robustness in detecting multiple fault types that demonstrate RF handles feature selection and classification simultaneously, which helps to improve reliability for condition-based maintenance systems.

A statistical features-based approach for bearing fault diagnosis using vibration single was proposed by Altaf et al. [4], in time, frequency and spectral power domains. They also extracted features like RMS, mean, etc. from both time domain and second derivatives and power spectral density and classified faults into healthy, outer race, inner race, and ball faults using KNN, SVM, Kernel LDA, by this they achieved high accuracy i.e. 99.13% with KLDA and 96.64% with KNN that reduces computational burden by 95%.

Fault detection system for rotating machines using vibrational signal analysis and EMD for denoising was developed by Kafeel et al. [5] and extracted multi domain features from 3D vibration signals representing healthy and faulty states, then after they classified faults using SVM, KNN, Decision Tree and LDA, testing multiple kernel settings. By this they achieved highest accuracy approximately 98% with SVM and Gaussian kernel along with 96% sensitivity and 100% specificity.

The bearing fault diagnosis using naturally occurring defects with vibration data processed with via SNR enhancement was investigated by Sahu et al. [6] in which they extracted 14 statistical features including RMS to represent fault patterns before classification and applied SVM, RF, LR, KNN and MLP to compare performance across machine learning models where they found RF consistently achieved the highest accuracy on feature enhanced data.

A deep groove ball bearing fault diagnosis was studied by Jaiswal et al. [7] using ResNet50, AlexNet50, ANN, decision Tree and Discriminant analysis by extracting 14 statistical features and they applied feature ranking method to evaluate feature importance by then classified faults into healthy, case fault, ball fault, inner ring fault and outer ring fault. They achieved 97.9% accuracy with ResNet50 and 97% with ANN.

Early fault detection in bearings was investigated by Ali et al. [8] using the time domain vibration analysis technique. They created the faults on the inner raceway and ball with size from 0.007” to 0.028”; then they extracted features like RMS, standard deviation and skewness which helped them to find difference between healthy and faulty bearings. Thei results showed that the faulty bearings have much higher vibration values which makes it easy to detect fault easy and maintenance.

A real time online fault diagnosis method was proposed by Wang et al. [9] for rolling bearings using K-Nearest Neighbor (KNN) algorithm and preprocessing vibration signals through data cleaning, segmentation and feature extraction before model training. Then conducted real time online fault diagnosis by continuously extracting features like RMS from incoming vibrational signals in which KNN outperformed C4.5 and CART classifiers, showing higher accuracy plus sustainability for rolling bearing monitoring.

Rolling element bearing fault diagnosis was investigated by Jamil et al. [10] using SVM and KNN classifiers trained on time and frequency domain features. They used CWRU open-source bearing data set for training and testing the models and found that frequency-domain features provide more reliable training for Ml models where KNN outperformed SVM in accuracy for detecting different bearing faults.

The multiple machine learning and deep learning techniques are compared by Farooq et al. [11] for predictive maintenance of ball bearing systems and they used a dataset 2155 vibration signals to train and evaluate the models based on accuracy, precision, recall, etc. In this extreme gradient boost achieved the best balance between high accuracy and low computation time.

3 Methodology

3.1 Dataset Overview

The dataset for this study is designed to give a reliable source of vibration information for evaluating the performance of machine learning models i.e. RF, SVM, KNN and LR based only on the single statistical feature - Root Mean Square. We took the experimental data from a laboratory test rig which is specially built for the bearing fault diagnosis studies. From that test rig we recorded the vibration responses of SKF 6206 deep groove ball bearing (DGBB) under two different health conditions, one is healthy and another is 0.5 mm defect at outer race.

The defect on the bearing was created artificially having 0.5 mm width and 1 mm deep groove on the outer race of the bearing which was produced by using the wire cut electro-discharge machining (EDM) that simulates the real-life bearing defects conditions with precision. The data was acquired at two motor speeds i.e. 1500 rpm and 1800 rpm, under no load and zero tilt or misalignment conditions to avoid the external disturbances. The wide range of sampling frequency allows accurate detection of even minor vibration variations that are caused due to the fault. This experimental configuration gives a highly stable and reproducible environment for isolating fault induced vibrations. With the help of these operating conditions, the study aimed to detect early fault caused by faulty bearing using minimal statistical features with improved efficiency.

3.2 Experimental Framework

In this study, we explored two main methods which were used for introduction of defects. The first method involves running the bearing until natural wear occurs, monitoring the gradual changes in vibration signals over time. The problem is that this approach effectively reflects the real-life conditions but highly consumes time and also, it's hard to control experimentally. Another second approach the most common method which includes artificially creating the faults on bearing. For precise control over the fault's size, shape and the location we used an electro-discharging machine (EDM). It helps to study the behavior of vibration under specific fault conditions effectively. In our study, we fabricated the defect on the outer race of the bearing, which almost stimulate the real-world fault conditions. That defect is of o.5 mm in width and 1 mm in depth, and it is compared with the healthy bearing conditions. Below in Table I, bearing faults and experimental conditions like load, tilt, device etc. is mentioned.

Table 1. Bearing Faults and Experimental conditions

Sr. No.	Parameter	Specifications
1.	Bearing Condition	Healthy, Outer Race Defect (0.5 mm)
2.	Rotational Speeds	1500 RPM, 1800 RPM
3.	Load	No Load
4.	Shaft Alignment	0° tilt

5.	Measurement Device	Tri-axial Piezoelectric Accelerometer
6.	Sampling Frequency	2560 Hz



Fig. 1. Healthy Bearing



Fig. 2. Outer Race fault in the SKF6206 Bearing with a 0.5 mm defct

The experimental setup was designed to provide a stable and controlled environment for collecting vibration data. The rig consisted of a rotor supported on two pedestals, with one end connected to a DC motor via a flexible coupling to drive the system and the other end serving as the measurement point. The assembly was mounted on a steel frame anchored to a concrete base to minimize external vibrations. The motor speed was adjustable, and experiments were conducted at 1500 rpm and 1800 rpm under no-load conditions to clearly observe the effects of defects. In this study, to capture the vibration signals we installed a high-sensitivity tri-axial piezoelectric accelerometer (Deltron type 2524-B) on the bearing pedestal in the radial direction The signals were recorded by the help of data acquisition system with a sampling frequency of 2560 Hz, then we analyzed them using FFT-based software after the system stabilized for approximately 15 minutes. Specifically, SKF 6206 deep groove ball bearings were used in our study and the precise creation of fault results in high reproducibility and reliability in the results. The vibration patterns generated from these analyses are highly characteristic and depend on both bearing and rotational speed (RPM). These patterns help to identify the type and location of faults effectively. We focused on extracting statistical features from the vibration signals rather than depending upon complex frequency-domain models which provides a simpler and efficient method for fault detection.

Table 2. Geometric details of SKF 6206 Deep Groove Ball Bearing

Sr. No.	Parameters	Value
1.	Bore Diameter (d)	30 mm
2.	Outside Diameter (D)	62 mm
3.	Outer Ring Shoulder Diameter (D1)	52.08 mm
4.	Inner Ring Raceway Diameter (F)	37.5 mm
5.	Pitch Diameter (dp)	46.48 mm
6.	Bearing Width (B)	16

7.	Ball Diameter (dr)	9.5 mm
8.	Number of Balls	9
9.	Defect Location	Outer Race
10.	Defect Dimensions	0.5 mm (width) × 1 mm (depth)
11.	Operational Speeds	1500 RPM, 1800 RPM
12.	Net Weight	0.19 kg
13.	Contact Angle	0 deg.

Our setup was designed to compare the vibration behavior of healthy bearings with the defective bearing. This allowed us to clearly observe the difference between vibration amplitude, frequency spectrum, and sidebands generated with the fault. By systematically analyzing these vibration features, we were able to accurately identify faults at an early stage and understand how the vibration characteristics evolve under different operational conditions.

3.3 Feature Extraction-Root Mean square (RMS)

Definition

The Root Mean Square (RMS) is a statistical measure that shows the effective or average power of a vibration signal. It shows the energy level of the signal by considering both positive and negative amplitudes. Instead of just focusing on the highest or lowest peaks, RMS provides an overall sense of how strong the vibrations are over time.

In other words, it converts the continuously changing vibration waveform into a single meaningful value that gives the overall vibration intensity. A higher RMS value shows the stronger vibrations, which can be a sign of defects.

Mathematically;

$$\text{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2} \quad (1)$$

where; x_n - Vibration amplitude at the n-th sample and N - The total number of samples.

Purpose for Selecting the RMS Feature

In vibration-based fault diagnosis, the RMS is considered as one of the most widely used statistical features. It directly represents the energy content of the vibration signal, with which we can reliably identify the mechanical health of bearings. Under normal circumstances, healthy bearings show low and stable RMS values, whereas faulty bearings give higher RMS values due to increased vibrations because of surface defects or cracks. One of the main motivations of using RMS is its simplicity and faster computation. It can be easily calculated from time domain data without using complex mathematical transformations such as Fourier or wavelet analysis which makes it highly suitable for real-time or embedded condition monitoring systems used in industries.

Moreover, several previous studies have verified the effectiveness of RMS as a strong and consistent indicator for bearing fault detection. Researchers like Tu et al.

(2023) and Altaf et al. (2022) reported over 95% accuracy when using RMS as the primary feature. Therefore, in our study, RMS was selected to evaluate the performance of different machine learning models in classifying bearing conditions based on their vibration energy levels.

RMS Calculations

The vibration signals were collected using a tri-axial piezoelectric accelerometer fixed on the bearing pedestal as discussed earlier. The data was recorded at a sampling rate of 2560 Hz for two conditions; healthy and faulty. The faulty condition was made by creating a 0.5 mm outer race defect with wire-cut EDM for accurate and repeatable results.

After collecting the signals, we divided into them smaller time parts to observe small changes in the vibration pattern. For each part, the RMS value was calculated using the given formula above. This RMS value shows the overall strength or energy of the vibration and each calculated RMS value was saved as one data sample, and the samples were labeled as 0 for healthy and 1 for faulty to make the dataset. When the RMS values of both conditions were compared, we found that faulty bearings always had higher RMS values which shows that RMS is a strong and simple feature that can help in finding and classifying bearing faults using machine learning methods.

Practical Insights into the RMS

The Root Mean Square (RMS) value plays an important role in understanding how rotating machines behave. It shows the total vibration energy that passes through the bearing. When a defect forms on the ball, inner race, or outer race, it creates small repeated impacts every time the rolling element moves over the damaged area. These impacts increase the vibration amplitude, which also increases the RMS value. So, if the RMS value keeps rising, then it usually means that a fault is starting or getting worse inside the bearing.

Because of its clear physical meaning, RMS becomes a simple yet powerful indicator of machine health. It allows maintenance engineers to quickly understand the condition of the machine without doing any hard mathematical calculations. Another big advantage is that RMS values can be measured continuously, which makes them very useful for real-time or online condition monitoring systems. It is also easy to calculate, works well even with small sets of data, and gives stable results under different machine speeds and loads.

However, RMS also has some limitations like even though it helps to detect faults, it cannot show the exact location or type of defect, such as whether it is on the inner race, outer race, or ball. Also, when more than one type of fault produces similar vibration levels, their RMS values can look the same, which can reduce accuracy. To solve this problem, RMS is often used together with other statistical features like Kurtosis, Skewness, or Crest Factor. Combining these features gives a completer and more accurate model for detecting and diagnosing faults in bearings.

Additional Statistical Features Considered

Beyond the Root Mean Square (RMS) value, the investigation rigorously incorporates several other critical statistical features to achieve a complete, multi-dimensional representation of bearing vibration behavior. These features are indispensable in vibration analysis because they each capture unique, non-overlapping characteristics of the signal's distribution. Specifically, the four primary metrics utilized are: Peak Value, which directly reflects the magnitude of sudden impact forces; Crest Factor, which is crucial for precisely indicating the sharpness and magnitude of high-energy impulses; Skewness, representing the asymmetry in the overall vibration distribution; and, most notably, Kurtosis, which is highly effective for highlighting severe impulsive, shock-type defects characteristic of early-stage spalling. Incorporating this broader set of metrics is essential. This multi-feature strategy significantly strengthens the overall fault classification process and provides a deeper, more robust understanding of the bearing dynamics during mechanical degradation.

Machine Learning Models for Fault Classifications

After finding the RMS feature from the vibration signals, different machine learning algorithms were used to check whether the bearing is healthy or faulty. The main idea of using machine learning is to let the computer automatically learn the hidden connection between the RMS values and the bearing's health condition. Once these models are trained properly, they can predict the condition of a new bearing using only its vibration data.

To make the results fair and trustworthy, four common supervised learning algorithms were selected — Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Logistic Regression (LR). Each of these algorithms learns in a different way, which helps to compare their performance and understand how well each model can classify the bearing condition using only the RMS feature.

Random Forest

The Random Forest (RF) algorithm is a type of ensemble learning model, which means it combines many small decision trees to make one strong and accurate final decision. Each tree is trained on a random part of the dataset, and the final result is decided by taking the majority vote from all the trees. This method helps to reduce errors and prevents overfitting, which means the model does not just memorize the data but learns general patterns.

In this study, the Random Forest model was built using 100 trees, each with a maximum depth of 10. This algorithm works well with vibration data because it can handle noise, non-linear patterns, and changes in signal behavior. Another advantage is that RF can show feature importance, which tells how much the RMS feature helps in making the classification. Random Forest was chosen because of its high accuracy, strong performance, and stability. In most bearing fault detection studies, it gives better and more reliable results than single machine learning models, even when the machine runs at different speeds and loads.

Support Vector Machine

The Support Vector Machine (SVM) is a powerful machine learning algorithm that tries to draw the best possible line or boundary, i.e. a hyperplane, between the two classes; healthy and faulty. It does this by keeping the biggest possible gap (margin) between the classes, which helps the model make more confident and accurate predictions.

In our study, the RBF (Radial Basis Function) kernel was used because it can handle non-linear data very well. The RBF kernel changes the simple RMS data into a higher-dimensional space, which makes it easier for SVM to separate the two classes even when the difference between them is very small. SVM was selected for this work because it gives high precision, works well for vibration and fault detection problems, and is mathematically stable. It is also less likely to overfit, meaning it performs well not only on training data but also on new, unseen data.

Additionally, SVM can efficiently handle high-dimensional datasets where other algorithms may struggle. Its ability to maintain clear decision boundaries even with noisy data makes it a strong and dependable model for bearing fault classification tasks.

K-Nearest Neighbors

The K-Nearest Neighbors (KNN) algorithm is a simple but easy-to-understand model. It predicts the class of a new data point by checking the K nearest samples (neighbors) in the training dataset. In this study, K = 5 was selected, and the Euclidean distance formula was used to measure how close the points are:

$$\text{Distance} = \sqrt{(\sum (x_i - y_i))} \tag{2}$$

Here, x_i and y_i represent the feature values of the test and training samples.

KNN is a non-parametric method, which means it does not assume any fixed pattern or relationship in the data. It only depends on the similarity between samples, making it flexible for vibration data where RMS values may slightly change. The main advantage of KNN is its simplicity and clarity — it is easy to understand and visualize how the algorithm makes predictions. However, it can sometimes be slow for large datasets and is sensitive to data scaling, so normalization was applied before training to improve performance. KNN performs well when the data distribution is simple and clear. It also serves as a good baseline model to compare with more complex algorithms like Random Forest or SVM in vibration-based fault detection.

Logistic Regression

The Logistic Regression (LR) algorithm is a linear model used for binary classification problems. It predicts the probability that a data sample belongs to a particular class; in our case, whether the bearing is healthy (0) or faulty (1). Even though it is a simple model, Logistic Regression gives a good baseline for comparing performance with more advanced algorithms.

It uses a sigmoid function to convert the RMS feature into a probability value between 0 and 1, as shown below:

$$P = \frac{1}{1 + e^{-(b_0 + b_1 X)}} \quad (3)$$

Where; Probability of a faulty condition b_0 is Intercept$_1$ is model coefficient and X is RMS feature.

Logistic Regression was selected to observe how a basic linear model performs compared to non-linear models like SVM and Random Forest. It is easy to implement, simple to understand, and requires very little computational power, which makes it suitable for real-time or embedded vibration monitoring systems. Additionally, it provides clear interpretation of model parameters, helping to understand how changes in RMS directly affect the fault probability.

In the present work, Logistic Regression is strictly treated as a classification technique but not as a linear regression method. Logistic Regression predicts the probability of a point to have any reserved attribute (i.e. "faulty" or "healthy") using the sigmoid function. This is a simple binary classifier and serve as baseline for comparing with non-linear classifiers (for example SVM, RF, KNN). The language in the explanation is also modified to Logistic Regression being a classification method.

4 Model Training and Testing

4.1 Data Preparation and Preprocessing

The experimental setup involved dividing the bearing fault dataset into separate training and testing subsets to ensure proper model evaluation. Along with RMS, additional statistical indicators such as Peak Value, Crest Factor, Skewness, and Kurtosis were also examined to represent bearing vibration more accurately. The dataset contained vibration signals from rolling element bearings under different health conditions, with RMS (Root Mean Square) values serving as the primary feature for fault detection.

Before model training, several preprocessing steps were applied to the data. Feature scaling was performed using Standard-Scaler to normalize the input features, which is particularly important for distance-based algorithms like KNN and SVM. The RMS values were extracted from vibration signals along with other operational parameters including speed, load, tilt angle, frequency, and defect size. The target variable was binary, classifying bearings as either healthy (0) or faulty (1).

4.2 Model Implementation

In this study, four machine learning models were used to compare their performance. Random Forest model had 100 trees with maximum depth of 10 to avoid overfitting but still keep good accuracy. It can also deal with non-linear relations, so it is useful for finding faults from vibration signals.

SVM model used RBF kernel with regularization value C = 1.0. By using kernel tricks, it could learn complex patterns after changing data into higher space. KNN used Euclidean distance and 5 neighbors after trying different values. This method worked well for finding similar vibration patterns in data. Logistic Regression was used as a

base model with L2 regularization to stop overfitting. It gave simple and easy results that helped for comparing other models.

4.3 Hyperparameter Selection

Selecting the optimal hyperparameters for all implemented models was achieved through a systematic process of trial-and-error experiments performed exclusively on the training dataset. This iterative approach was critical to ensuring both high accuracy and consistent, reproducible model performance. For the K-Nearest Neighbors (KNN) model, numerous K values were evaluated, with the final stable configuration determined to be K=5. Similarly, the Support Vector Machine (SVM) required testing multiple kernel types; the Radial Basis Function (RBF) kernel was ultimately chosen, paired with a regularization parameter of C=10, as this combination yielded the best classification accuracy. When configuring the Random Forest classifier, the number of decision trees was strategically varied; the combination of n-estimators = 100 and a maximum tree depth (max depth) of 10 proved highly effective, balancing strong performance with the avoidance of overfitting. Finally, since the initial analysis utilizes only a single statistical feature (RMS), the Logistic Regression model was implemented using its default L2 regularization setting, which provides sufficient control over the minimal feature space. These carefully chosen settings guarantee robust and reliable results across the entire suite of models.

4.4 Training Procedure

A systematic approach was followed in the training process for all algorithms. The dataset was divided in an 80–20 ratio, where 80% of the samples were used for training and 20% for testing. This division ensured that there was enough data for learning while still keeping sufficient samples for evaluation. During training, each algorithm learned how the input features, such as RMS values and operational parameters, were related to the target labels representing the bearing health status. To minimize the loss functions, Random Forest used Gini impurity, SVM used hinge loss, and Logistic Regression used log loss. Cross-validation was not applied because the dataset size was small, but the same train-test split across all algorithms helped in making a fair comparison. The training continued until each model reached its convergence point.

While the current methodology relies on a fixed 80–20 train–test split for baseline consistency, future enhancements will rigorously incorporate k-fold cross-validation (specifically 5-fold or 10-fold) to significantly improve reliability. This addition is crucial: it effectively mitigates the performance bias inherently associated with a single fixed split. Cross-validation guarantees that every single data sample is utilized for both training and testing across various iterative folds. This practice will provide a far more stable and trustworthy estimate of model performance, ultimately strengthening the generalization capability and overall robustness of the reported diagnostic results.

4.5 Testing and Evaluation

The model was tested using an unseen dataset to check its generalization capability. Predictions were made for each test sample, and the model's performance was evaluated using multiple metrics.

Accuracy shows how correct the model's predictions are. It is found by taking the ratio of correct predictions to the total number of samples. Precision tells how well the model avoids false alarms in fault detection, while recall shows how many actual faults were correctly identified. The F1-score gives a balanced value between precision and recall to represent overall performance.

Confusion matrices were made for every algorithm to show true positives, true negatives, false positives, and false negatives. These matrices helped to study the error patterns and to understand how each model behaved under different fault conditions.

Using the same evaluation method for all algorithms made the comparison fair and helped to understand the strengths and weaknesses of each model in industrial condition monitoring.

4.6 Performance Comparison

The experimental results showed different performance levels for the four algorithms. Random Forest gave the highest accuracy in bearing fault detection because its ensemble method captured complex vibration patterns. SVM also performed well, especially in dealing with non-linear decision boundaries in the feature space. KNN achieved reasonable accuracy with its instance-based learning approach, while Logistic Regression acted as the baseline model. The comparison helped to understand the balance between model complexity, computation time, and detection accuracy in real bearing fault diagnosis cases.

5 Results of Classification Metrics for Different Models

5.1 Comparison pf Classification Metrics for Different Methods

Table 3. Performance Comparison of Machine Learning Models Using RMS Feature

Sr. No.	Model	Accuracy (%)	Precision (%)	Sensitivity (%)	F1(%)
1.	RF	95	75	100	85
2.	SVM	90.91	100	50	66.66
3.	KNN	88.89	100	33.33	50
4.	LR	72.73	60	42.86	50

The above Table 3 shows the comparison of different machine learning models based on the RMS feature. From these results, it was observed that the Random Forest performed the best with an accuracy of about 95%. It was able to correctly identify all

faulty cases, as indicated by 100% sensitivity. The model also maintained a good balance between precision and recall, which is shown by the F1 score of 85%.

The SVM was able to achieve 90.91% accuracy with 100% precision, but its sensitivity was only 50%, which shows that it missed some fault cases even though the correct predictions were accurate. The K-Nearest Neighbor (KNN) model gave 88.89% accuracy with high precision (100%) but very low sensitivity (33.33%), showing that it correctly identified a few faulty samples. The Linear Regression (LR) model performed the lowest among all, with an accuracy of 72.73%, which means it is not suitable for fault detection using this dataset.

From these results, it can be concluded that the Random Forest model gives the most reliable performance for detecting bearing faults using the RMS feature. Its ability to handle complex and non-linear relationships makes it more effective for predictive maintenance applications.

5.2 Evaluation of Machine Learning Algorithms with RMS Feature

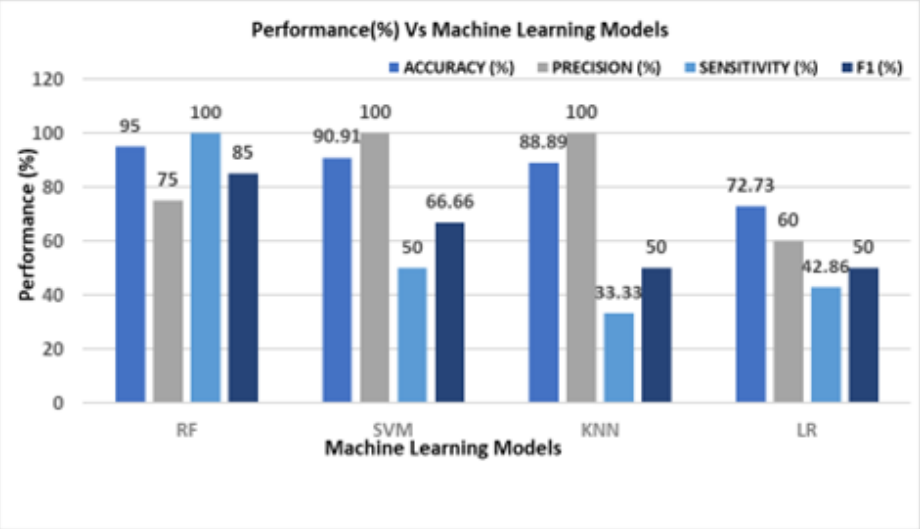


Fig. 3. Performance Comparison of Machine Learning Models Using RMS Feature

Fig 3 shows the performance of different machine learning algorithms was studied using the RMS feature extracted from vibration signals. The RMS value mainly represents the overall energy level of the signal and helps to identify variations caused by bearing faults. Based on the observations from the results, the following behavior of each algorithm was noticed.

The Random Forest (RF) model gave the most reliable results among all. It was able to clearly separate healthy and faulty conditions because of its ability to handle complex

and non-linear data patterns. Since RF is based on multiple decision trees, it reduced the chances of error and improved the overall prediction consistency.

The Support Vector Machine (SVM) also performed well. It could correctly classify most of the data samples, but it was slightly less effective in detecting all fault types. This is because RMS alone may not capture subtle vibration differences, and SVM relies on class margins. The K-Nearest Neighbor (KNN) algorithm presented a mixture of performance. It detected the major fault patterns accurately but struggled when the values of features for various conditions were near each other. As KNN works on the distance between data points, the RMS feature sometimes gave overlapping results, which affected its fault detection ability.

The Logistic Regression (LR) model gave comparatively lower performance. This is because LR is a linear model and cannot properly fit the non-linear relations present in vibration data. The RMS feature by itself does not provide enough information for a linear model to draw an accurate decision boundary.

Overall, it was observed that algorithms capable of learning non-linear relationships, such as RF and SVM, perform better with the RMS feature. Models that depend only on linear separation, like LR, or distance-based comparison, like KNN, show limited performance. Hence, RMS can be considered an effective but insufficient feature alone for fault classification, and combining it with other statistical parameters can give more accurate results.

5.3 Statistical Analysis of the Proposed Work

A statistical analysis was performed to evaluate the performance of the proposed machine learning models. Standard statistical metrics such as accuracy, precision, sensitivity, and F1-score were calculated to measure the classification capability of each model. The confusion matrix for every algorithm was also analyzed to study the distribution of true positives, true negatives, false positives, and false negatives. These statistical indicators provide a clear understanding of the reliability and consistency of the classifiers. The Random Forest model achieved the highest accuracy and sensitivity, indicating strong fault-detection capability. The SVM and KNN models showed high precision but lower sensitivity, which suggests they were more conservative in predicting faults. Logistic Regression demonstrated weaker statistical performance due to the linear nature of the model. Overall, these statistical results validate the effectiveness of the proposed RMS-based approach in differentiating healthy and faulty bearings

In the presence of noise, varying sampling frequencies, or real-time streaming, the classification performance may degrade if no preprocessing is applied. Industrial environments introduce mechanical noise, electromagnetic interference, and speed fluctuations. To maintain performance, preprocessing techniques such as band-pass filtering, wavelet denoising, resampling, normalization, and segmentation would be required. Random Forest is generally more noise-tolerant, whereas SVM and KNN are more sensitive without filtering. For real-time deployment, fixed-length windowing and online feature extraction (RMS, Kurtosis, bandwidth energy) would be required along with a low-latency classifier. Integrating these techniques would allow the proposed framework to operate reliably in practical industrial applications.

6 Conclusion and Future Work

6.1 Conclusion

In this study, we used a machine learning approach to identify faults in rolling element bearings using vibration signals. Random Forest, Support Vector Machine (SVM), K-Nearest Neighbor (KNN), and Logistic Regression were the four algorithms tested based on the RMS feature. In this, Random Forest (RF) gives the highest accuracy, and SVM also performs well. The RMS feature proved to be a useful indicator because it reflects the overall vibration energy and is simple to compute. This study shows that machine learning works better than conventional methods for identifying bearing faults at an early stage. For future work, adding more features and testing in real situations would make it even better. Using machine learning helps keep machines running safely and prevents unexpected breakdowns.

6.2 Future Work

In future, this work can be made better by adding more statistical features and trying other machine learning methods to see how they perform. It can also be used in real-time by taking data from bearings running at different speeds and loads. Using more data and adjusting the model settings may help to improve the accuracy and reliability of the results.

Performance enhancement hinges critically on incorporating more discriminative features than solely relying on the RMS value. Why is this necessary? Because RMS only provides the total energy of the vibration, it fundamentally misses the critical, impulsive details. Therefore, a richer feature space is absolutely necessary. This includes the highly sensitive time-domain features—Kurtosis, Crest Factor, and Peak Value—which are essential for capturing the shock-related information produced uniquely by localized defects. Furthermore, the framework requires frequency-domain features, such as band-energy metrics, to specifically target and detect characteristic fault frequencies and their associated sidebands. By combining this robust set of metrics, a significantly more informative feature space is established. This allows sophisticated algorithms like Random Forest (RF) and SVM to form dramatically clearer decision boundaries, ultimately resulting in improved accuracy, enhanced sensitivity, and better robustness across differing defect severities. Rigorous exploration of this crucial enhancement is planned for future work, extending the capability of the current single-feature RMS approach.

While the present work establishes a strong baseline, it is crucial to acknowledge its current scope: a focus only on a single defect size (0.5 mm), a no-load condition, and purely binary classification (healthy vs. faulty). Real industrial systems present far greater complexity. Varying defect sizes, dynamic rotational loads, and multi-fault combinations dramatically alter the vibration signatures. Although larger defects predictably increase amplitude and simplify model separability, small or early-stage defects present a far greater classification challenge. Moreover, fluctuations in load and

RPM fundamentally change the signal's frequency content, which demands high generalization capability from the models. Due to their inherent robustness against noise and non-linear patterns, algorithms like Random Forest and SVM are typically better suited for handling such variations than KNN. Therefore, a necessary future extension involves rigorously broadening the framework to include multiple defect sizes, varying operational loads, and multi-fault scenarios to evaluate the model's true generalization capacity in realistic industry environments.

The reliability of classification may degrade significantly if no preprocessing is applied, especially when dealing with noise, varying sampling frequencies, or real-time data streaming. Industrial environments are inherently complex, introducing mechanical noise, electromagnetic interference, and inevitable speed fluctuations that compromise signal quality. To maintain high performance in these conditions, preprocessing techniques such as band-pass filtering, wavelet denoising, resampling, normalization, and segmentation would be absolutely required. It is important to note that Random Forest is generally more tolerant of noise than SVM and KNN, which are significantly more sensitive without explicit filtering. For practical, real-time industrial deployment, the framework would require implementing fixed-length windowing and online feature extraction (RMS, Kurtosis, bandwidth energy), paired with a low-latency classifier. Integrating these robust techniques will allow the proposed framework to operate reliably and effectively in demanding, practical industrial applications.

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7.2 Disclosure of Interests

The authors of the paper “Data-Driven Fault Detection in Rolling Element Bearings Using Machine Learning for Predictive Maintenance” affirm that they have no conflict of interest to disclose for the publication of this research paper. The research was performed for academic and research purposes only. The authors affirm that no external financial support or funding was provided for the study that could influence the outcomes and/or conclusions of this study. All authors approved of the final manuscript for publication.

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